Big Effects from Small Samples:

Album Releases and Traffic Fatalities

Gaurav Sood* Daniel Weitzel†

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Abstract

Patel et al. (2026) report that major album releases raise US traffic fatalities by 18.2 deaths, or 15.1%, and read the increase as evidence of smartphone distraction. The estimate replicates. Applying their specification to an independently assembled Fatality Analysis Reporting System extract gives +17.6 deaths ($SE = 7.45$, clustered by album). The explanation that suggests itself first, that nine of the ten releases fell on a Friday and Fridays are dangerous, does not survive: ten Fridays drawn at random reach the observed effect 3 times in ten thousand, the mean Friday residual after the paper's fixed effects is a seventh of a death, and a model-free contrast of each release against its own neighbouring Fridays gives +14.1, a ratio the original authors also report. Applying the estimator to the same calendar positions in other years returns -0.08.

What does not survive is the magnitude. Ten treated days cannot detect an effect below 13.4 deaths at eighty percent power, and at the paper's own clustered standard error the threshold is 23.1, larger than the estimate reported. Across the 27 major releases that are not the paper's own, the pooled effect is +6.5 with a ninety-five percent interval of [-0.2, 13.1]. A true effect of that size, filtered through significance at a standard error of 7.45, produces published estimates averaging 17.6. The reported eighteen is not in tension with a true effect near six; it is what a true effect near six looks like when it reaches print.

The sharpest evidence on magnitude comes from measuring the dose. The paper's first-day stream counts are global while its outcome is American, so we rebuild the daily Spotify Charts top two hundred for the United States and get a dose measured the way the outcome is, along with a comparison set defined by a rule rather than a list. 17 albums other than the paper's reached its own smallest release, 32.1 million measured first-day US streams. Their release days average -0.49 deaths ($SE = 2.54$) against +16.1 for the ten, and -1.24 when restricted to the paper's own years. Across all 524 charting albums with a measurable first day, release size does not predict release-day fatalities.

The mechanism remains untested. The comparison of crashes with and without an impaired driver now covers all ten albums, using NHTSA's multiply imputed blood alcohol files rather than the `DRUNK_DR` field that leaves the crash record after 2020, but it splits on an outcome and

*gsood07@gmail.com

†weitzeld@ceu.edu

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so identifies nothing. The alternative that would most threaten the causal reading, that a busy week generates both more streaming and more driving, is testable once the streaming series is in hand, and it fails: ambient listening, the daily US chart total with that day's new releases removed, does not predict fatalities at all, and adding it moves the estimate by two hundredths of a death. The anomaly is real, it is not a day-of-week artifact, and it is not ambient activity. It is a property of those ten albums rather than of releases their size.

1 Introduction

Distracted driving is a serious and well-documented road-safety problem. Naturalistic driving studies show that visual and manual distraction substantially elevate crash risk, and the transportation-safety literature has long distinguished distraction and inattention from other forms of driver impairment (Klauer et al., 2006; Regan et al., 2011; Dingus et al., 2016). Regulatory guidance reflects the same concern: the National Highway Traffic Safety Administration has issued formal recommendations aimed at reducing visual-manual distraction from in-vehicle electronic devices (National Highway Traffic Safety Administration, 2013). Against that backdrop, the question raised by Patel et al. (2026) is both plausible and important: can a recurring cultural event—the release of a major music album—generate a measurable spike in fatal crashes by increasing smartphone-based streaming while people are driving?

Patel et al. (2026) answer yes. Using the Fatality Analysis Reporting System (FARS) and a sample of ten highly streamed albums released between 2017 and 2022, they report that fatalities rise by 18.2 deaths, or 15.1%, on release days. That claim attracted wide public attention because it links a familiar and recurring consumer event to a large mortality effect. It also speaks to a broader empirical literature that uses narrow-window event studies to infer causal effects from shocks tied to particular calendar days.

We replicate the central statistical pattern closely. Applying the paper's specification to our independently assembled FARS extract yields an estimate of 17.6 excess deaths on release days ($SE = 7.45$, clustered by album), very near the reported 18.2.

The obvious objection does not survive. Nine of the ten release days are Fridays, and Fridays are high-fatality days in FARS, so the estimate looks like it could be a day-of-week artifact. It is not. Ten Fridays drawn at random reach the observed effect 3 times in ten thousand; the comparison is unchanged by whether weekday fixed effects are in the model at all; letting the Friday effect vary by season and by year moves the estimate by about a death; a contrast of each release against its own neighbouring Fridays, which uses no model, recovers a ratio the original authors also report; and applying the estimator to the same calendar positions in other years returns -0.08. The day -6 coefficient that looks like a pre-trend is a Saturday, and the actual previous Friday is flat.

What does not survive is the magnitude. Ten treated days cannot detect an effect below 13.4 deaths at eighty percent power, and at the paper's own clustered standard error the threshold is 23.1, larger than the estimate reported. Across the 27 major releases that are not the paper's own, the pooled effect is +6.5 with a ninety-five percent interval of [-0.2, 13.1]. A true effect of that size, filtered through significance at a standard error of 7.45, produces published estimates averaging 17.6. The reported eighteen is not in tension with a true effect near six; it is what a true effect near six looks like when it reaches print.

These concerns place the paper in a familiar methodological terrain. Several papers in adjacent settings have argued that narrow-window comparisons around salient calendar days can produce striking estimates that weaken under broader or better-matched comparisons. Zhang and Aronow (2016) revisit an earlier claim about fatal crashes on U.S. presidential election days; Palayew and Harper (2019) do the same for the annual cannabis holiday; and Brown et al. (2018) raise similar concerns in a reanalysis of the "4/20" crash claim. The contribution here is to apply the same diagnostic logic to the music-streaming hypothesis while also recognizing that some features of the original result are persistent and require explanation.

Two distinct questions run through this debate. One is whether smartphone distraction raises crash risk. The broader literature already gives a clear answer on that point (Klauer et al., 2006; Regan et al., 2011; Dingus et al., 2016; National Highway Traffic Safety Administration, 2013). The other is whether this design identifies a release-day streaming effect in national fatalities. The analysis below bears on the second question. It should not be read as casting doubt on the general danger of distracted driving.

Album releases really do cluster on Fridays, largely because of the industry's global release convention, so the day-of-week concentration is not a consequence of researcher discretion. What limits the claim is arithmetic rather than identification. The original paper detects a real anomaly on a small set of release dates. Its size is the part the design cannot support, and the mechanism is a separate question that none of the available tests can settle.

Table 1: Regression estimator. Paper's specification on stacked album-days: day-of-week, week-of-year, year and holiday fixed effects, standard errors clustered by album (G = 10, so use t with 9 df).

estimator	sample	effect	se	pct_effect	n_albums	source
Paper (reported Figure 2B)	Tier 1 (2018-2022)	18.20	5.50	15.10	10	Patel et al. (2026)
Paper spec (our replication)	Tier 1 (2018-2022)	17.61	7.45	14.45	10	This analysis
Paper spec (full data)	Tier 1 (all years)	17.61	7.45	14.45	10	This analysis
Paper spec (pre-2018)	Tier 0 (2015-2017)	12.61	5.51	13.03	10	This analysis
Paper spec (out-of-sample)	Tier 3 (2023-2024)	-7.98	7.56	-7.26	7	This analysis
Paper spec (all tiers)	All 37 albums	10.58	3.25	9.65	37	This analysis

2 Replication

Using FARS data for 2007–2024 and the ten Tier 1 albums listed in Patel et al. (2026), we estimate a linear regression of daily national fatalities on a release-day indicator, with day-of-week, week-of-year, year, and federal-holiday fixed effects, within a ±10 day event window around each release. Standard errors are clustered by album, reflecting that the treatment varies across only ten units.

The reported and replicated estimates are close. Patel et al. report +18.2 deaths. We estimate +17.6 deaths ($SE = 7.45$), a 14.4% increase on a control-day mean of 121.9 fatalities, which is the analogue of the 120.9 they report. The point estimates differ by 0.6 deaths, well within one standard error.

Our wider standard error reflects album-level clustering, which is the natural adjustment when treatment varies across ten units. That also sets the critical value: with ten clusters the interval belongs on nine degrees of freedom, giving [0.8, 34.5] and $p = 0.042$. The replication is significant at conventional levels and not comfortably so, which is the first sign of the sample-size problem developed in Section 5.

Randomization inference confirms that the anomaly is not merely a tail event in the residual distribution. Drawing 10,000 placebo treatment vectors under four schemes—random ten days, stratified nine-Friday-one-Sunday, Friday-only, and seven-day block bootstrap—returns $p < 0.01$ in all four. Studentized and album-clustered variants return $p = 0.01$. The central question is what this pattern represents.

3 What the Estimate Is Not

Before turning to what the design cannot show, three properties of the estimate rule out the simplest ways of dismissing it.

Not a single-specification artifact. We estimate 35 specifications varying event-window width (± 5 to ± 21 days), sample period (pre-2018, 2018–2022, all years), and album set (Tier 1, Tier 1+2, all 37 albums). All return the same sign and 30 return $p < 0.05$. The count of 35 overstates the independence of these tests: they are seven distinct album-set and sample-period configurations run at five window widths, and widening the window from ± 5 to ± 21 days moves the Tier 1 estimate by under two deaths. Read as configurations rather than as tests, six of seven are significant and the single null is the pre-2018 sample.

Not a narrow-window artifact. A standard critique in this literature is that widening the comparison window dissolves the effect. That failure mode does not apply here, as the window rows above show.

Concentrated on the release day. The dynamic event-study shows the largest coefficient on day 0 (+16.1, $t = 3.04$), with smaller coefficients on adjacent days. The day-1 estimate of +8.8 is not distinguishable from zero once the critical value reflects the ten albums the standard error is computed from ($t = 1.94$, 95% CI $[-1.4, +19.0]$).

Friday releases are an industry convention. All nine Friday releases follow the industry's global Friday release convention, under which major commercial releases in the United States overwhelmingly land on Fridays. The day-of-week concentration is a structural feature of the treatment rather than evidence of opportunistic date selection.

4 What Does Not Explain It

Nine of the ten Tier 1 albums were released on a Friday and the tenth on a Sunday. Fridays average 117.6 fatalities in FARS 2017–2022 against an overall daily mean of 107.5, and ninety percent of

Table 2: Residual estimator across 7 album-set and sample-period configurations at 5 window widths. The 5 nulls are one configuration repeated, so read 6 of 7 configurations rather than 30 of 35 tests.

window	album_set	sample_period	n_albums	effect	se	t_stat	p_value
5	Tier1	2018-2022	10	14.78	5.45	2.71	0.02
5	Tier1	all_years	10	15.95	5.30	3.01	0.01
5	Tier1+2	2018-2022	20	14.15	4.17	3.39	0.00
5	Tier1+2	all_years	20	14.83	4.16	3.57	0.00
5	All	pre-2018	10	5.08	4.53	1.12	0.29
5	All	2018-2022	21	13.63	4.00	3.41	0.00
5	All	all_years	37	9.21	2.84	3.25	0.00
7	Tier1	2018-2022	10	14.65	5.47	2.68	0.03
7	Tier1	all_years	10	15.99	5.31	3.01	0.01
7	Tier1+2	2018-2022	20	14.24	4.22	3.38	0.00
7	Tier1+2	all_years	20	15.00	4.19	3.58	0.00
7	All	pre-2018	10	4.80	4.51	1.06	0.32
7	All	2018-2022	21	13.82	4.05	3.41	0.00
7	All	all_years	37	9.20	2.85	3.22	0.00
10	Tier1	2018-2022	10	14.82	5.45	2.72	0.02
10	Tier1	all_years	10	16.10	5.30	3.04	0.01
10	Tier1+2	2018-2022	20	14.50	4.25	3.41	0.00
10	Tier1+2	all_years	20	15.07	4.20	3.59	0.00
10	All	pre-2018	10	4.68	4.52	1.04	0.33
10	All	2018-2022	21	14.05	4.07	3.45	0.00
10	All	all_years	37	9.14	2.86	3.20	0.00
14	Tier1	2018-2022	10	14.91	5.40	2.76	0.02
14	Tier1	all_years	10	16.19	5.28	3.07	0.01
14	Tier1+2	2018-2022	20	14.89	4.30	3.46	0.00
14	Tier1+2	all_years	20	15.20	4.22	3.60	0.00
14	All	pre-2018	10	4.80	4.50	1.07	0.31
14	All	2018-2022	21	14.05	4.08	3.44	0.00
14	All	all_years	37	9.14	2.86	3.19	0.00
21	Tier1	2018-2022	10	14.29	5.35	2.67	0.03
21	Tier1	all_years	10	15.90	5.27	3.02	0.01
21	Tier1+2	2018-2022	20	14.93	4.38	3.41	0.00
21	Tier1+2	all_years	20	15.27	4.25	3.59	0.00
21	All	pre-2018	10	4.41	4.50	0.98	0.35
21	All	2018-2022	21	14.20	4.16	3.41	0.00
21	All	all_years	37	8.94	2.89	3.10	0.00

Table 3: Residual estimator with the additive day-of-week term replaced by the listed interaction. mean_friday_resid shows how little additive Friday signal the baseline leaves.

model	n_params	effect	se	t_stat	p_value	mean_friday_resid
DOW + month + year (baseline)	37	16.10	5.30	3.04	0.01	0.14
DOW x year	139	15.95	5.37	2.97	0.02	0.13
DOW x month	103	16.24	5.80	2.80	0.02	0.15
DOW x year + DOW x month	211	16.16	5.88	2.75	0.02	0.14
DOW x week-of-year	390	15.07	4.73	3.18	0.01	0.12

release days are Fridays against twelve percent of the days in their own control windows. The imbalance is as large as it looks. It is also not the explanation.

4.1 Day of Week

The most direct test uses no model. For each Friday release we compare the release day with the same album's own Fridays seven and fourteen days either side, which holds weekday, season and year fixed by construction. Pooled across the nine Friday releases that gives +14.1 deaths, a ratio of 1.12, positive for seven of nine. Patel et al. (2026) run the same comparison and report a ratio of 1.10. It is also imprecise: 95% CI [-2.4, 30.6], $p = 0.084$. Nine treated units do not buy significance for an effect this size, a point we return to in Section 5.

The model-based version is sharper. Drawing ten Fridays that are not release days and running them through the same estimator ten thousand times, 3 draws reach the observed +16.1. Patel et al. (2026) report twenty exceedances in a thousand iterations of their own version; the two agree.

That comparison does not depend on the weekday fixed effects, which is worth checking rather than assuming. Removing them shifts the mean Friday residual from 0.15 to 9.25 and the observed effect from 16.0 to 25.4, leaving the contrast between them at 15.9 and 16.2 and the placebo p-value unchanged. Dummies for a weekday add a constant to every day of that weekday, and a Friday-against-Friday comparison differences it away.

What a residual-based placebo cannot see is whether release Fridays are unusual *kinds* of Fridays, since season and year are absorbed. Replacing the additive weekday term with interactions answers that. Across day-of-week interacted with year, with month, with both, and with week of year, from 37 to 390 parameters, the estimate stays between 15.07 and 16.2, and every specification clears $p = 0.022$.

The event study appears at first to say otherwise, with a coefficient of +16.5 on day −6, nearly the size of the release-day coefficient. Day −6 is not the previous Friday. For nine of the ten albums it is the Saturday before release. The previous Friday is day −7, at +3.8 ($t = 0.92$), and the following Friday is day +7, at −0.3. Both same-weekday neighbours are quiet, which is the adjacent-Friday contrast read off the event study. A joint test of flatness across the ten pre-treatment days, flipping the sign of each album's whole residual vector so that the correlation induced by using the same ten albums for every coefficient is respected, returns $p = 0.19$; dropping day −6 returns $p = 0.70$.

4.2 Calendar Position

Labels choose release dates, so the dates are not random. If the choice favours positions that are also high-risk, the estimate picks that up. Applying each album's date to every other year in the series, matched to the nearest same weekday, reproduces the seasonal and holiday context without the album. Those 168 placebo dates average -0.08 deaths (SE 1.48, clustered by album, since they come from ten calendar positions rather than 168 independent draws). Dates falling inside a real release window are excluded.

Two of the ten releases sit within three days of a federal holiday. Widening the holiday control from the release day itself out to seven days either side moves the estimate between 15.4 and 15.9, and dropping both albums leaves 15.6 on the remaining 8.

4.3 One Unusual Day

With ten treated units the mean is hostage to any one of them, and *Her Loss* sits far out at +56.8. Statistics that a single large day cannot move agree with the mean. The median release-day residual is 12.9 and the twenty percent trimmed mean 13.0. Nine of the ten release days sit above their prediction, which a sign test puts at $p = 0.021$, and a signed-rank test at $p = 0.0039$. Dropping *Her Loss* leaves 11.6 ($p = 0.006$), with a tighter interval than the full sample, since that album contributes most of the variance as well as most of the excess.

Weather on the specific dates does not account for it either. Snow and blowing snow make up under 1.4% of fatal crashes on every one of the ten days, and the wettest release day relative to its own window has one of the smaller residuals. Those figures are shares of the crashes that happened rather than station readings, so they describe the crashes and cannot serve as controls; the station series of Section 7 does that job.

5 What the Design Can and Cannot Show

The residual standard deviation of daily national fatalities after these fixed effects is 13.7 deaths, so a ten-day mean has a standard error of 4.33. With ten treated units the critical value is t on nine degrees of freedom, and the smallest effect the design detects eighty percent of the time is 13.4 deaths. Measured against the dispersion actually observed across the ten releases, 5.30, the threshold rises further. At the paper's own album-clustered standard error of 7.45 it is 23.1, larger than the estimate the paper reports.

se_source	se	mde_80	mde_50	true_effect	power	exaggeration	wrong_sign	n_al
null (residual SD / sqrt n)	4.33	13.44	9.80	0.00	0.05	NaN	NaN	
null (residual SD / sqrt n)	4.33	13.44	9.80	2.00	0.07	4.78	0.13	
null (residual SD / sqrt n)	4.33	13.44	9.80	3.00	0.10	3.30	0.05	
null (residual SD / sqrt n)	4.33	13.44	9.80	5.00	0.18	2.12	0.01	
null (residual SD / sqrt n)	4.33	13.44	9.80	8.00	0.38	1.48	0.00	
null (residual SD / sqrt n)	4.33	13.44	9.80	12.00	0.69	1.16	0.00	
null (residual SD / sqrt n)	4.33	13.44	9.80	16.00	0.91	1.04	0.00	
null (residual SD / sqrt n)	4.33	13.44	9.80	20.00	0.98	1.01	0.00	
null (residual SD / sqrt n)	4.33	13.44	9.80	16.10	0.91	1.04	0.00	
realized across albums	5.30	16.44	11.98	0.00	0.05	NaN	NaN	
realized across albums	5.30	16.44	11.98	2.00	0.06	5.75	0.17	
realized across albums	5.30	16.44	11.98	3.00	0.08	3.96	0.08	
realized across albums	5.30	16.44	11.98	5.00	0.14	2.52	0.02	
realized across albums	5.30	16.44	11.98	8.00	0.27	1.72	0.00	
realized across albums	5.30	16.44	11.98	12.00	0.53	1.30	0.00	
realized across albums	5.30	16.44	11.98	16.00	0.77	1.12	0.00	
realized across albums	5.30	16.44	11.98	20.00	0.92	1.04	0.00	
realized across albums	5.30	16.44	11.98	16.10	0.77	1.11	0.00	
paper's clustered SE	7.45	23.12	16.85	0.00	0.05	NaN	NaN	
paper's clustered SE	7.45	23.12	16.85	2.00	0.06	8.00	0.25	
paper's clustered SE	7.45	23.12	16.85	3.00	0.07	5.43	0.16	
paper's clustered SE	7.45	23.12	16.85	5.00	0.09	3.40	0.05	
paper's clustered SE	7.45	23.12	16.85	8.00	0.16	2.26	0.01	
paper's clustered SE	7.45	23.12	16.85	12.00	0.30	1.63	0.00	
paper's clustered SE	7.45	23.12	16.85	16.00	0.48	1.34	0.00	
paper's clustered SE	7.45	23.12	16.85	20.00	0.67	1.18	0.00	
paper's clustered SE	7.45	23.12	16.85	16.10	0.49	1.34	0.00	

Table 4: Residual estimator, mean per-album effect by album tier. Comparable to +16.1 for Tier 1, not to the regression estimate of +17.6.

tier	n_albums	n_positive	sign_test_p	avg_effect	se	t_stat	ci_lower	ci_upper
Tier 0 (pre-2018)	10	6	0.75	6.40	4.70	1.36	-4.23	17.02
Tier 1 (paper)	10	9	0.02	16.10	5.30	3.04	4.12	28.09
Tier 2 (extended)	10	7	0.34	13.05	6.60	1.98	-1.87	27.98
Tier 3 (post-paper)	7	3	1.00	-2.76	3.09	-0.89	-10.32	4.81
All combined	37	25	0.05	9.09	2.81	3.23	3.38	14.79
Excluding the paper's 10	27	16	0.44	6.49	3.23	2.01	-0.16	13.13

Table 5: What a ten-album study would report, given a true effect and the paper's album-clustered standard error of 7.45. Reconciles the pooled estimate outside the paper's sample with its headline.

true_effect	se	power	mean_published	p_reaches_replicated
6.49	7.45	0.12	17.58	0.46
9.09	7.45	0.19	18.46	0.55
16.10	7.45	0.49	21.51	0.76

An estimator can be unbiased and its published estimates still run high. Unbiasedness holds across every sample the design could have produced; what reaches print is the subset that cleared significance, and when power is low that subset is the large draws (Gelman and Carlin, 2014). If the true effect were five deaths, this design reaches significance 0.18 of the time and reports 2.1 times too much when it does.

5.1 Evidence From Outside the Ten Dates

Locating the magnitude requires evidence the ten dates cannot supply. We assembled 37 major releases from 2015 through 2024 and applied the same estimator to each.

Pooling all 37 includes the ten albums under examination and so cannot serve as evidence from outside them. On the 27 remaining albums the estimate is +6.5 with a ninety-five percent interval of [-0.2, 13.1], and 16 of 27 are positive, which a sign test cannot separate from a coin ($p = 0.44$).

That reconciles the two numbers. At a standard error of 7.45, a true effect of 6.5 reaches significance 0.12 of the time and the surviving estimates average 17.6, with 0.46 of them at or above the 17.6 we replicate.

5.2 The Mechanism Is Untested

Three tests bear on the proposed channel and none of them can carry it.

The sober-versus-drunk comparison is the paper's strongest mechanism evidence, and it runs on all ten albums. `DRUNK_DR` leaves the accident file after 2020, which would restrict it to two, but NHTSA's multiply-imputed driver BAC ships in every annual archive, and combining the ten imputations by Rubin's rules gives +13.07 deaths ($SE = 4.31$, $p = 0.003$) in crashes with no driver at or above the legal limit against +3.04 ($SE = 2.58$, $p = 0.242$) in crashes with one. The two sum to the all-crash effect.

That is a decomposition rather than a mechanism test, and the distinction is not pedantic. Whether the extra crashes involve a drinking driver is itself an outcome, so the split conditions on a post-treatment characteristic, in the same family as the crash-composition weather shares. It establishes where the excess sits. A concentration among sober crashes is consistent with distraction, and equally consistent with any mechanism that concentrates ordinary daytime exposure.

Album-level effects do not rise with first-day streams: across the twenty albums for which counts exist, $r = -0.17$, 95% CI [-0.57, 0.30]. At twenty albums a correlation must exceed 0.44 to reach $p < 0.05$, so this is a reading from an instrument with no resolution at this range rather than a wrong-signed finding. Eight of the twenty counts are estimated from chart position rather than measured, which attenuates any true relationship; the albums are all large, so the range of the dose is narrow; and first-day streams is a level where the mechanism needs an increment over the artist's ordinary streaming.

Applying the estimator to variables that describe the crashes rather than counting them turns up one result. Mean crash latitude is +0.37 degrees on release days ($t = 2.91$). This is not a placebo failure, because latitude summarises the same crashes whose count is the outcome and is therefore not independent of treatment. Neither is it demonstrably a composition artifact: if the extra crashes dragged the mean, albums with the largest excess would show the largest shift, and the correlation across the ten is -0.23 ($p = 0.52$), with *Her Loss* carrying the largest excess and shifting latitude by 0.029 degrees. Ten albums cannot settle it. With seven composition variables tested it is also the kind of result seven tests produce one of.

6 Streaming Volume, Measured

Two questions the fatality data cannot answer on its own need a streaming series: whether a busy week generates both more listening and more driving, and whether the effect tracks the size of the release. Both are answerable with the product Patel et al. (2026) themselves use, the daily Spotify Charts top two hundred for the United States, which also replaces a hand-assembled album panel with a rule.

The reconstruction matches theirs on the control days and not on the release days. Over the ten days surrounding each release the paper reports 86.1 million top-200 US streams and we get 86.8, with an interval of $[84.7, 88.8]$ that covers their figure. On the ten release days themselves the paper reports 123.3 million and we get 133.2, eight percent higher. We cannot account for the gap. It does not propagate, because release-day streaming volume is a consequence of the release and nothing downstream conditions on it, but it is the one place our series and theirs disagree and it should be on the record.

6.1 Ambient Listening Does Not Explain It

The alternative worth taking seriously is that a week in which more people are out and about generates both more streaming and more driving. Total streaming cannot test it, because total streaming on a release day is caused by the release; conditioning on it would absorb the mechanism rather than the confounder. Background streaming can: the top-200 total with every track released that day removed, which is ambient listening the release did not cause.

It does not predict fatalities. Regressed on the same fixed effects with release windows excluded, the coefficient is -0.0074 deaths per million streams ($SE = 0.0266$), so a one standard deviation move in ambient listening, 18.8 million streams, is -0.14 of a death. Adding it as a control moves the release-day estimate from $+15.42$ to $+15.40$.

6.2 The Effect Belongs to the Ten Albums, Not to Releases That Size

Every album on the US daily chart whose measured first-day streams reach the smallest of the paper's ten shows no release-day excess, unless it is one of the paper's ten. There are 17 such albums and they average -0.49 deaths ($SE = 2.54$, 95% CI $[-5.87, 4.89]$), against $+16.1$ for the

Table 6: Background streaming is the daily US top-200 total with every track released that day removed, so it is ambient listening the release did not cause. The total-streaming row is a bad control shown for contrast, not a robustness check.

specification	n_albums	effect	se	ci_lower	ci_upper	p_val
no streaming control	10	15.415	5.444	3.101	27.730	0.0
background streaming	10	15.401	5.453	3.066	27.737	0.0
total streaming (BAD CONTROL, shown for contrast)	10	15.947	5.447	3.624	28.269	0.0

ten. The comparison set is a rule rather than a judgement about which releases were major: reach 32.1 million first-day US streams and you are in it.

Getting to that comparison took two corrections to the paper's dose. The first is that its dose is global while its outcome is American. Table 1 of Patel et al. (2026) credits *Midnights* with 184.7 million first-day streams against 79.6 million on the US chart, and *Un Verano Sin Ti* with 145.8 million against 32.1. The larger figures are the widely reported global first-day totals. A US dose belongs with a US fatality count, and the paper's own streaming series is American.

The second is that ten albums cannot support a dose-response test. A correlation on ten units has to exceed 0.63 to register, which is most of the range the statistic can take, so the wrong-signed -0.50 the original albums produce carries almost no information. Every album on the chart with a measurable first day and a fatality residual gives 524 of them, where the threshold falls to 0.086.

At that resolution there is no dose-response to find. The correlation is 0.068 ($p = 0.120$) and the slope is 0.0927 deaths per million, but Figure 1 shows that the slope is not describing a trend. Twenty equal-count bins of albums the paper did not study, spanning two orders of magnitude of dose, all sit within five deaths of zero, and so does the bin above the paper's own cutoff. What looks like a dose-response in the pooled decile means is the paper's ten albums occupying the top decile. Restricting the comparison albums to the paper's own years leaves it where it was, at -1.24 on 9 albums.

The reading that survives is the one the out-of-sample albums already suggested by a different route. Release size does not predict release-day fatalities. Being one of the ten albums a published study selected does.

Table 7: Holding measured dose fixed. Every album on the US daily chart whose first-day streams reach the smallest of the paper's ten is a comparison unit, which makes the comparison set a rule rather than a judgement about which releases were major. Effects are means of donut residuals with standard errors across albums.

group	n	effect	se	ci_lower	ci_upper	p_value
the paper's ten	10	16.10	5.30	4.12	28.09	0.01
every other album reaching 32.1M	17	-0.49	2.54	-5.87	4.89	0.85
the same, 2018-2022 only	9	-1.24	2.04	-5.94	3.46	0.56
albums below 32.1M	497	0.73	0.68	-0.61	2.07	0.29

Table 8: Release-day excess by decile of measured first-day US streams, across every album on the chart. The ten-album version of this test could not detect a correlation below 0.63; this one detects above 0.086.

dose_bin	n	dose_median	effect_mean	effect_se
0	53	0.971	2.112	1.576
1	52	1.401	2.466	2.061
2	52	1.958	0.395	2.007
3	53	2.688	-3.441	2.299
4	52	3.734	-0.680	2.054
5	52	4.927	2.422	2.089
6	53	6.399	0.872	2.146
7	52	8.663	1.864	2.598
8	52	13.747	-0.718	2.185
9	53	32.096	4.538	1.891

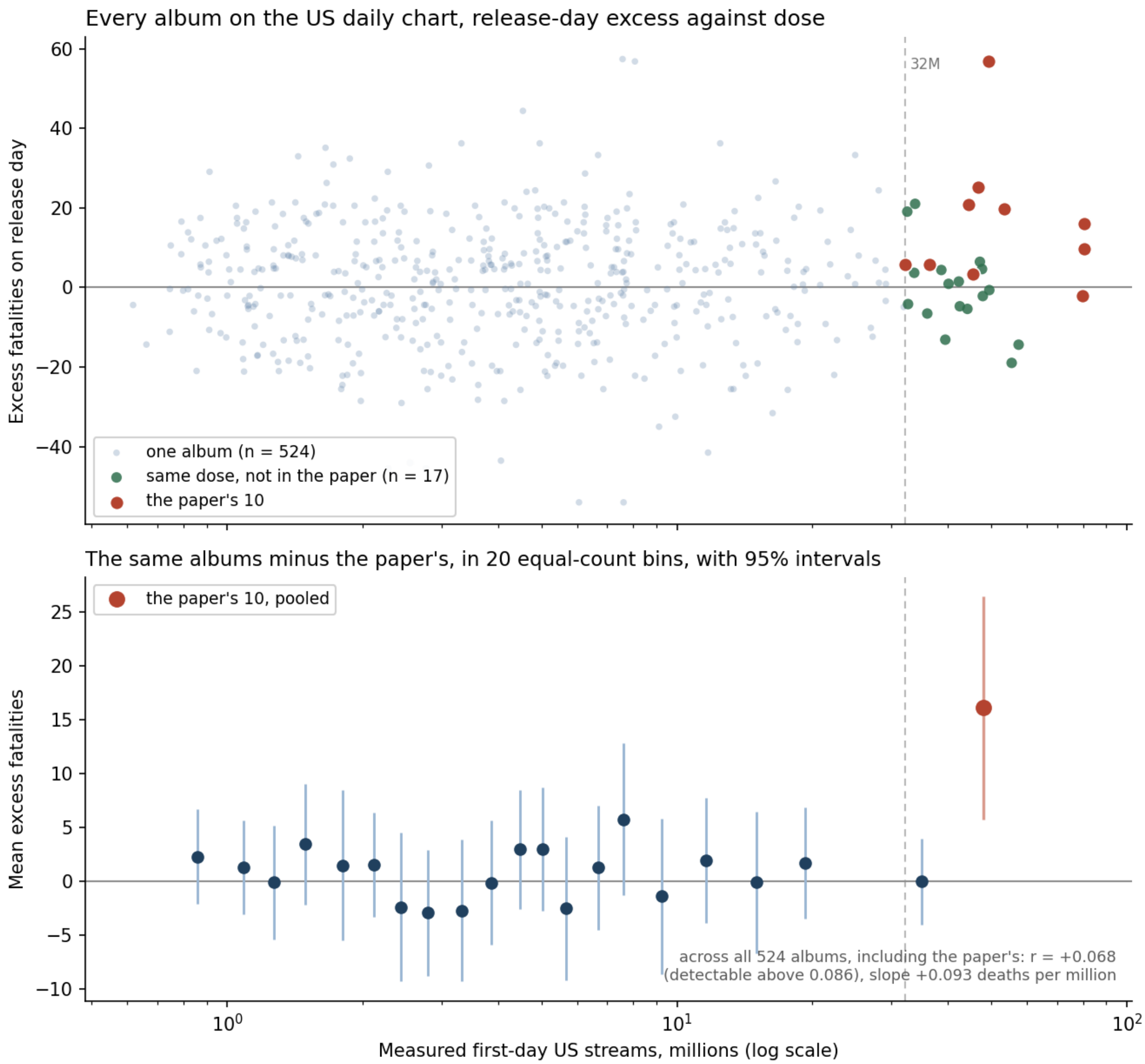

Figure 1: Release-day excess fatalities against measured first-day US streams, for every album on the daily Spotify top 200 between 2017 and 2024 with at least three charting tracks released on the same day and a fatality residual available ($n = 524$). Excess is the residual from the donut model of Section 2: observed release-day fatalities minus the prediction of a day-of-week, month, year and holiday model fitted with release windows excluded. The dashed line marks 32.1 million streams, the smallest of the ten albums in Patel et al. (2026); red points are those ten and green points are the 17 other albums that reach the same dose. The lower panel drops the paper's ten and groups the rest into twenty equal-count bins on the same logarithmic axis, with bin means and 95% intervals from the spread of albums within a bin; the red point is the paper's ten pooled, plotted at their median dose. The horizontal line is zero excess.

Table 9: Albums 2018-2022 ranked by measured first-day US streams from Spotify Charts, so the sample is generated by a rule rather than assembled by hand. The estimate is present at every frame size and falls as the frame widens.

frame_size	n_albums	min_first_day_millions	effect	se	ci_lower	ci_upper	p_value
10	10	45.408	13.242	5.688	0.374	26.110	0.045
15	15	36.341	10.078	4.254	0.953	19.203	0.033
20	20	31.912	7.248	3.405	0.122	14.373	0.047
30	30	21.908	6.361	2.843	0.547	12.176	0.033
50	50	15.539	4.313	2.055	0.183	8.443	0.041

Table 10: Release-day effect with daily precipitation from NOAA stations in all 51 states, weighted by each state's share of US fatalities. Unlike the crash-composition shares, nothing here is a function of the crash record.

specification	n_controls	effect	se	ci_lower	ci_upper	p_value
no weather control	0	14.0333	5.2411	2.1771	25.8895	0.0253
precipitation, fatality-weighted	1	14.2374	5.2287	2.4093	26.0655	0.0235
precipitation, equally weighted	1	14.5388	5.2938	2.5633	26.5142	0.0226
both weightings together	2	14.5067	5.2848	2.5517	26.4617	0.0227

6.3 A Frame Anyone Can Rebuild

Ranking every album released between 2018 and 2022 by measured first-day US streams replaces a hand-assembled list with a rule. The estimate is present at every frame size and declines monotonically as the frame widens, from +13.24 on the top ten to +4.31 on the top fifty.

That decline is what selection on the largest releases looks like when it is measured rather than argued. The effect is real at every cut, and its magnitude is largest where the sample is most extreme, which is the same conclusion the out-of-sample albums reached by a different route.

7 Limitations of This Critique

Weather enters through daily precipitation from NOAA GHCN stations in all fifty states and the District of Columbia, aggregated with fixed weights equal to each state's share of US traffic fatalities from 2007 to 2024, so the control weights by where fatal crashes happen and nothing in it is a function of the crash record. Adding it moves the estimate from 14.03 to 14.24, a spread of 0.51 deaths across weightings.

That series has three limits. One station per state under-represents weather inside large states.

It covers precipitation only: GHCN precipitation includes the water equivalent of snow, but snowfall as a separate variable is absent, because GSOD carries snow depth rather than snowfall amount. And five states use their principal airport rather than a downtown station, since downtown co-operative records are too sporadic.

The weather variables that appear in the crash-composition tables are shares of the fatal crashes that occurred in rain, fog, cloud or snow, computed from the crash file itself. Numerator and denominator are both the outcome, so the share falls mechanically when release days add crashes, which makes them inadmissible as controls (Angrist and Pischke, 2009). They are reported as description of what the extra crashes looked like and nothing conditions on them.

The measured dose is censored from below. The chart covers the top two hundred songs, so an album contributes only its charting tracks and an album that places three tracks is measured less completely than one that places twenty. The censoring applies to Patel et al. (2026) equally, since its streaming series is the same product. It also bounds coverage in time: the US daily chart begins in 2017, so the 37-album panel still rests on a hand-assembled list for earlier releases, and the reproducible frame of Section 6 runs from 2018 forward.

The alcohol decomposition splits crashes on a characteristic the release itself may change. It locates the excess and cannot identify what produced it, and no comparison built from the crash record alone can, because the crash record does not say who was listening to anything.

With ten treated units in the original sample and 37 in the broader panel, all inference here is fragile, and that cuts both ways. The out-of-sample estimate has an interval that touches zero. The 2023–2024 subset that looks like a break rests on seven albums. Small-sample instability affects this critique exactly as it affects the paper. The dose-matched contrast is the one comparison here that does not depend on a small treated group for its comparison side, and its 17 comparison albums are still few enough that its interval spans five deaths in each direction.

8 Conclusion

The anomaly on the original ten dates is real, it replicates closely, and it is not an artifact of the day of the week. Random Fridays do not reproduce it, the comparison is invariant to whether weekday dummies are in the model, letting Friday vary by season and by year leaves it intact, a model-free contrast against each album's own neighbouring Fridays recovers it, and the calendar positions carry nothing in other years. The day -6 coefficient that looks like a pre-trend is a Saturday, and the actual previous Friday is flat.

What does not hold is the size. Ten treated days cannot detect an effect below 13.4 deaths, and at the paper's own standard error the threshold exceeds the estimate it reports. Outside the ten chosen dates the pooled effect is +6.5 with an interval that touches zero, and a true effect of that size, filtered through significance, produces published estimates averaging 17.6. The anomaly on those ten dates is real; the effect of releasing an album is much smaller than the number attached to it.

Measuring the dose sharpens that from an interval to a comparison. The 17 albums whose first day on the US chart was at least as large as the paper's smallest release show no release-day excess at all, -0.49 deaths against +16.1 for the ten. Whatever produced the anomaly, it was not the size of the release.

The mechanism is a separate question and remains open. The comparison that most directly supports distraction covers all ten albums, but it splits crashes on a characteristic the release itself may change, so it describes where the excess sits and cannot say why. The rival explanation that would matter most, ambient activity, is answerable with public data and does not hold.

The broader lesson is not about music. Narrow-window designs built on a handful of salient dates have produced striking estimates before, on election days (Zhang and Aronow, 2016) and on the cannabis holiday (Palayew and Harper, 2019; Brown et al., 2018), and the reanalyses have tended to find smaller effects rather than none. A design that can only see large effects will only ever report large ones. The remedy is more treated units, and here they were sitting in the same data product the original paper used to measure its treatment.

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